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NOTES ON NEGLECTED AND UNDERUTILIZED CROPS

Problems, progress and future prospects of improvement of *Commiphora wightii* (Arn.) Bhandari, an endangered herbal magic, through modern biotechnological tools: a review

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Abstract *Commiphora wightii* (Arn.) Bhandari syn. *C. mukul* Engl. (Burseraceae) is an economically and pharmacologically important slow growing, dioecious, balsamiferous woody, multipurpose shrub heading towards extinction. Commonly known as "Guggul" due to the presence of steroidal compound guggulsterone in the oleo-gum resin, it has been used in treating various ailments and disorders since ancient times (2000 B.C.). Evaluation and confirmation of hypolipidemic effects of guggul based on Ayurvedic text in 1960s provided a new insight into its pharmacological applications. Two bioactive isomers of guggulsterone, E and Z, are responsible for lipid- and cholesterol-lowering activities. Recently, it has been shown to have anti-cancerous activity also. It is found in the dry regions of Indian subcontinent, namely India, Pakistan and Bangladesh. Ruthless and unscientific harvesting of oleo-gum resin from the wild, by local populations, for economic benefits with negligible conservation efforts has made this species endangered and has led to its inclusion in Red Data Book of IUCN. Although this plant has many excellent traits, adequate attention has not been focused on its conservation and improvement. Conventional propagation methods i.e., seeds, cuttings and air layering are in place but have many limitations. Therefore, application of modern biotechnological tools needs to be standardized for harnessing maximum benefits from this pharmaceutically important plant. An efficient regeneration system needs to be in place for improvement of this genus through genetic transformation and production of useful metabolites in cell cultures. Studies are in progress for micropropagation through shoot multiplication and somatic embryogenesis, as well as for secondary metabolite (guggulsterone) production in callus cultures and bioreactors. No selected germplasm is available for *C. wightii* since it is a wild plant. Breeding programs have not yet been started due to lack of systematic cultivation and conservation programs. Moreover, little information has been gathered regarding the genetic variability in this species using RAPD and ISSR markers. No details are available about genetic makeup and QTL linkage maps. Investigations are in progress to search sex linked markers in this dioecious species. Research is

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also in progress to decipher the molecular mechanisms underlying various pharmacological actions of guggul. Since the approval of use of guggul as a food supplement by United States Food and Drug Administration in 1994, an exponential increase in research publications on various aspects of research on guggul have been published. Present communication summarizes the problems, progress made and suggests some future directions of research for this important endangered medicinal plant.

Keywords *Commiphora wightii* · Endangered medicinal plant · Guggulsterone · Micropropagation · Pharmaceutical applications · Secondary metabolite production

Introduction

Genus *Commiphora* (2n= 26) (Sobti and Singh 1961) commonly known as Guggul or Myrrh tree is a small tree of the family Burseraceae. Burseraceae, a torchwood family, is also known as frankincense family Sapindales, because of its association with aromatic and anti-inflammatory resin in vertical resin ducts and canals (Weeks and Simpson 2005). Genus *Commiphora* comprises of approximately 165 species of small thorny, sturdy, highly branched, balsamiferous trees with a short trunk up to 3–4 m tall with thin papery bark (Barve and Mehta 1993). The plant has crooked, knotty branches ending in sharp spines. The younger parts are pubescent and glandular leaves are trifoliate. The generic epithet *Commiphora* originates from the Greek words *kommi* (meaning 'gum') and *phero* (meaning 'to bear') as majority of the species yield a fragrant oleo-gum-resin following damage to the bark. *Commiphora* is a very slow growing plant adaptable to high temperature (45 °C) which prefers arid and semi-arid climates with annual rainfall of 225–500 mm and is also tolerant of poor soil. Various species of genus *Commiphora* are native to different countries of world including central Asia (Barve and Mehta 1993), India (Kant et al. 2010), Bangladesh and Pakistan (Deng 2007), Arabia (Oman and Yemen) as well as Somalia and Ethiopia in Northeast Africa (Dharmananda 2003). The best frankincense is found in Oman (Dharmananda 2003). Somalia is the major exporter of scented myrrh at present (Mabberley 2008). Approximately 150 species are distributed around the Red Sea of East Africa and 20 species in Madagascar (Hanuš et al. 2005). The distribution further extends to Australia and Pacific Islands (Lal and Kasera 2010). Siddiqui (2011) reported the presence of four species yielding guggul in India namely *C. wightii* (Arn.) Bhandari, *C. mukul* Engl., *C. agallocha* Engl. and *C. berryi* (Arn.) Engl. while Barve and Mehta (1993) have reported the occurrence of 5 species in India out of which *C. wightii* (Arn.) Bhandari and *C. roxburghii* Engl. yield guggul. On the other hand, Ramawat et al. (2008) reported that three species of *Commiphora* occur in India namely *C. wightii* (Arn.) Bhandari, *C. stocksiana* Engl. and *C. berryi* (Arn.) Engl. Atal et al. (1975) also reported the occurrence of three species in India. *C. wightii* is found in some districts of Rajasthan, Madhya Pradesh, Gujarat and Karnataka where it is used in Ayurvedic formulations while *C. agallocha* Engl. is distributed in Orissa and *C. berryi* (Arn.) Engl. regionally known as Mulkoluvai is used as hedge plant all over Southern India (Lal and Kasera 2010; Siddiqui 2011). Table 1 details the distribution and economic importance of some important *Commiphora* species.

Previously known as *Commiphora mukul* Engl. or *Balsamodendron mukul* Hook. ex Stock this plant was renamed *C. roxburghii* by Santapau in 1962. Later, Bhandari (1964) reported that the scientific name *wightii* was published in 1839 that is prior to *roxburghii* in 1948, hence, *Commiphora wightii* (Arn.) Bhandari is the correct and valid Latin name (Lal and Kasera 2010). *C. wightii* (Fig. 1a, b) is the most important species of this genus and has been used extensively in pharmaceutical and perfumery industry. It is native to India, Bangladesh and Pakistan (Deng 2007; Urizar and Moore 2003) and may be found from Northern Africa to Central Asia (Barve and Mehta 1993) but is most common in Northern India (Kant et al. 2010). In India it is found in arid, rocky and sandy tracts of Rajasthan, Gujarat, Maharashtra, and Karnataka states (Soni 2010a). This plant has been known by various names like Indian bdellium in English (Lal and Kasera 2010), guggul in Hindi (Murray 1995), gukkulu and mainshakshi in Tamil and guggulu in Sanskrit (Patil et al. 1972). This in fact gave the generic name to the plant, in English "Guggul" means Guggulsterone an active ingredient of this plant that was used to treat a variety of ailments in ancient Indian Ayurvedic system.

Table 1 Distribution and economic importance of some *Commiphora* species

Species	Distribution	Uses	Reference(s)
C. africana (A. Rich) Engl.	Sub Saharan Africa, Northern Nigeria	Antimicrobial effect, lowering serum cholesterol, thyrogenic effect ($T_4 \rightarrow T_3$)	Aliyu et al. (2002)
C. agallocha Engl.	India, W. Pakistan, Arabia, Tropical and Southern Africa	Grown along field bunts and boundaries	Yadav et al. (1999)
C. aprevali Guillaumin	Medagascar	Suitable for bonsai	Arid land greenhouses (2011)
C. berryi (Arn.) Engl.	Southern India	Used as hedge plant	Siddiqui (2011), Lal and Kasera (2010)
C. buruxa Swanepoel	Southern Namibia	Less than 20 plants known	Swanepoel (2011)
C. confusa Vollesen	Kenya	Planted in villages to serve as hedge or fence, leaves are browsed by livestock. In northern Kenya, the resin is used as incense	Dekebo et al. (2002)
C. edulis (Klotzsch) Engl.	South Africa	Active against the MCF-7 cells (SRB anticancer assay); used in traditional healing rites	Paraskeva et al. (2008)
C. erlangeriana Engl.	Ethopia, Somalia	Fruits are poisonous to humans and mammals, used as arrow poison	Dekebo et al. (2002), Habtemariam (2003)
C. erythraea Engl.	Kenya, Southern Arabia, Somalia, Eastern Ethopia	Has larvicidal and repellent activity against *Amblyomma americanum* (star tick) and *Dermacentor variabilis* (American dog tick). Is a source of opopanax	Carroll et al. (1989)
C. glandulosa Schinz	South Africa	Leaf and stem extracts are effective against MCF-7 cells, SF-268 cells and HT-29 cells (SRB anticancer assay); used in traditional healing	Paraskeva et al. (2008)
C. guidotti Chiov. ex Guid.	Somalia	Contains strong bacteriolytic compound T-cadinol	Claeson et al. (1992)
C. holtziana Engl.	Kenya	Used as a general acarine repellent, activity was confirmed against *Dermanyssus gallinae* (red poultry mite). Leaves are eaten for food	Birkett et al. (2008)
C. incisa Chiov.	Kenya	Mansumbinoic acid isolated from resin is quite effective in reducing joint swelling in adjuvant arthritis in rats	Duwiejua et al. (1993)
C. kataf Engl.	Kenya, Southern Arabia, Somalia, Eastern Ethopia	African opopanax	Baser et al. (2003)
C. kua (R.Br. ex Royle) Vollesen	Kenya, Ethopia and Somalia	Anti-inflammatory, used in incense	Awadh Ali et al. (2008)
C. marlothii Engl.	South Africa	Anti bacterial activity—kills Staphylococcus aureus; against the MCF-7 cells and HT-29 cells (SRB anticancer assay; used in traditional healing rites	Paraskeva et al. (2008)
C. mollis Engl.	South Africa	Anti-oxidant activity; used in traditional healing rites	Paraskeva et al. (2008)
C. molmol Engl. ex Tschirch	Kenya	Involved in wound healing mechanism by inducing maturation/differentiation/activation of both myeloid and lymphoid cells during the effector phase of the immune system	Haffor (2009)

Table 1 continued

Species	Distribution	Uses	Reference(s)
C. mukul Engl.	India	Cholesterol lowering effect, used to treat cardiac disorders	Verma et al. (2010), Nohr et al. (2009), Ojha et al. (2008)
C. myrrha Engl.	Arabia	Used in perfumery, remedy for throat and mouth ulcers	Hough et al. (1952)
C. neglecta I. Verd.	Kenya, Swaziland and Mozambique	Known as "sweet-root corkwood". Leaves and roots are eaten. Stem extracts exhibited good anti-oxidant activity in DPPH assay; used in traditional healing rites	Paraskeva et al. (2008)
C. opobalsamum Engl.	Southern Arabia and Abyssinia	Cardiovascular and hypotensive effect	Al-Howiriny et al. (2005)
C. ornifolia (Balf. f.) J. B. Gillett	Soqotra Archipelago of Yemen	Used to make naqf (from bark and under bark) to cure wounds and ulcers in both humans and livestock	Mothana et al. (2009)
C. parvifolia Engl., *C. planifrons* Engl., *C. socotrana* Engl.	Soqotra Archipelago of Yemen	Used to cure wounds and ulcers in both humans and livestock	Mothana et al. (2009)
C. pyracanthoides Engl.	South Africa	Contains many glycosides; Active against the MCF-7 cells and SF-268 cells (SRB anticancer assay); used in traditional healing rites	Paraskeva et al. (2008)
C. schimperi Engl.	South Africa	Active against the MCF-7 cells (SRB anticancer assay); used in traditional healing rites	Paraskeva et al. (2008)
C. stockiana Engl.	India	Always bisexual and polyembryonic. Contains Guggulsterone E and Z in minute quantities	Yadav et al. (1999)
C. tenuipetiolata Engl.	South Africa	Anti-oxidant activity; used in traditional healing rites	Paraskeva et al. (2008)
C. tenuis Vollesen	Ethopia	Young twig chewed for juice/water. Highly potent against *Staphylococcus aureus*	Asres et al. (1998)
C. viminea Burtt Davy	South Africa	Active against the MCF-7 cells (SRB anticancer assay); used in traditional healing rites	Paraskeva et al. (2008)
C. wightii (Arn.) Bhandari	India, Bangladesh and Pakistan	Oleo-gum resin contains guggulsterone E and Z responsible for cholesterol and lipid lowering properties	Satyavati et al. (1969)

The species is dimorphic with male, female and polygamous plants. The flowers may be uni- or bisexual, with the unisexual flowers only being semi-developed with non-functional stamens (Rao et al. 2001). Gender in guggul seedlings can be determined only during flowering. The tree drops its leaves during rainy season followed by flowering (October–December) and fruit set (October–January) in its natural range in India. The young leaves appear towards the end of the dry season. *C. wightii* has thin papery bark (Fig. 1c). The fruits, 6–8 mm in diameter have attractive colors, varying from pink to red. The fruit of *Commiphora* greatly enhances the identification of the species. When ripe, the fruit splits into halves revealing a brightly coloured pseudo-aril. The shape of the pseudo-aril differs from species to species.

India produces only 100 MT of guggul oleo-gum resin against its requirement of 1,000 MT. The deficit is met through imports from Pakistan and Afghanistan. India spends Rs 45 crore on its imports mainly from Afghanistan (Maheshwari 2010). The price of guggul oleo-gum resin over last 10–15 years have increased from Rs. 100–400 kg^{-1} indicating many fold increase in its use as well as decrease in available plants. It is

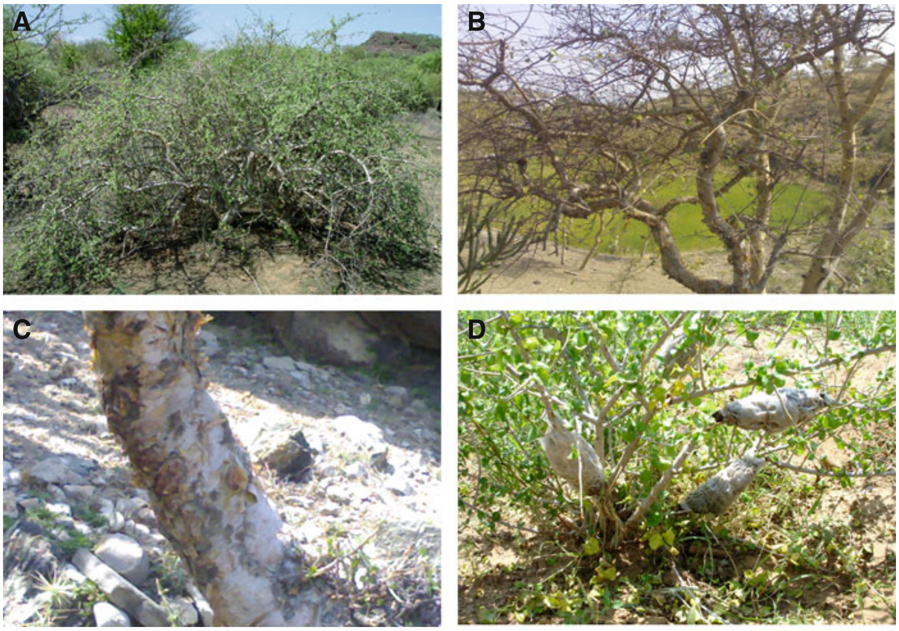

Fig. 1 *Commiphora wightii* **A** a green leafy tree, **B** leafless dormant tree, **C** stem with thin papery bark, and **D** propagation through air layering

considered as a threatened plant in India and has been included in Red Data Book of IUCN (IUCN 2011). Ruthless exploitation of this species has led to decline in its population continuously making this plant vulnerable which has been categorized as 'data deficient' in assemblage of IUCN in 2008 due to lack of research for establishment of its conservation status. Though, *C. wightii* is assigned to the DD (Data Deficient) category ver. 2.3 (1994) of the Red Data Book of IUCN, the Government of India has included it under RET (Rare, Endangered, Threatened) category (Samantaray et al. 2011). Currently only few wild populations exists in the states of Rajasthan and Gujarat. Community-based conservation drives to save this important medicinal plant have been initiated in Rajasthan, India (Soni et al. 2010b). Guggal Bachao Abhiyan (Save Guggul Movement) is being carried out in collaboration with villagers and tribals. For such type of conservation support has been received from the IUCN Sir Peter Scott Fund.

Economic importance

In addition to the multifarious pharmaceutical uses, it is also important for perfumery and incense industries (Barve and Mehta 1993). It is cherished by some multinational companies, like Unilever, and Proctor and Gamble, for preparation of soaps, detergents, creams, lotions and perfumes (Farah 1994). Oleo-gum resin exudates of *C. myrrha* Engl. are widely used in food industries for making beverages, chewing gums, candies, gelatins, nut products and confectioneries, puddings and canned vegetables due to some attractive properties viz good adhesive thickeners, emulsifier, fixatives, flavouring and perfect stabilizing agent (FAO 1995). Food and Drug Administration, USA approved the use of myrrh for food products (21 Code of Federal Registration-CFR 172.510), while the Council of Europe included myrrh in the list of plants and parts thereof that are acceptable for use in foods (Ford et al. 1992; Council of Europe 1981; Lemenih and Teketay 2003). In addition, it can be used as flies, mosquito and termite repellents after blending in incense sticks (Farah 1994).

Pharmaceutical importance

The multifarious medicinal and therapeutic values of *Commiphora* were known as early as 2000 B.C. in the *Atharva Veda*, one of the four well-known Ancient classic Scriptures (Vedas) of India (Ramawat et al. 2008; Shishodia et al. 2008). *Sushruta Samhita* a well-known Ayurvedic medical text describes the usefulness of the oleo-gum resin from the tree *C. wightii* in the treatment of a number of ailments, including obesity and disorders of lipid metabolism (Urizar and Moore 2003;

Nadkarni 1954). In biblical times, *C. myrrha* was valued as much as gold, and is mentioned numerous times in *The Old Testament*, in instructions to Moses about making incense and anointing oil, for which it has been used throughout history. Also of great religious importance, it was presented as one of the three gifts to Christ by the Magi, and was used to anoint the body of Christ after the crucifixion (Dharmananda 2003). The major commercial source of myrrh is *C. myrrha*, although the myrrh of the Bible is believed to be *C. guidottii* Chiov. ex Guid. *C. guidottii* was mentioned by Pliny as 'the scented myrrh,' and was used by the Romans as incense in temples (Mabberley 2008). It is said that Greek soldiers would not go into battle without a poultice of myrrh to apply as a dressing on their wounds (Hanuš et al. 2005). Today, nearly all human civilizations in the world (Indians, Greeks, Chinese Egyptians, Arabs and Romans) are benefited from this natural herbal remedy (Lemenih and Teketay 2003). The importance of this highly valuable medicinal plant was re-discovered by an Indian research student G.V. Satyavati in 1966 while confirming the medicinal properties of guggul based on ancient Ayurvedic text. She demonstrated the hypolipidemic effects of guggulsterone on rabbits. After successful experiments, this ayurvedic formulation was approved for marketing in India as a hypolipidemic drug (Deng 2007; Siddiqui 2011). Presently many ayurvedic formulations based on guggul are marketed in India like *Yograj Guggulu, Laksha Guggulu, Kaishore Guggulu, Gokshuradi Guggulu, Kanchanara Guggulu, Panchamrita louha Guggulu, Panchatikta Guggulu Ghrita, Punarnava Guggulu, Simghanada Guggulu, Trayodasanga Guggulu* and *Triphala Guggulu.* Among them, important herbal formulation like '*Laksha guggulu'* relieves pain resulting from dislocation of joints and bone fracture, and is also used to treat cardiac complaints, malnutrition and vatrog; '*Kaishore Guggulu'* another herbal medicine is used as dietary supplement for purification of blood and anti-allergic reactions. It is also beneficial in diabetes and arthritis (Lather et al. 2011). '*Yograj Guggulu'* is used in rheumatoid arthritis and for removal of toxins from body for maintenance of healthy metabolism. '*Triphala Guggulu'*, is especially useful in old age for the treatment of piles while '*Vatari Guggulu'* is used effectively against lower backache, rheumatism, gout and sciatica (Pradhan and Dash 2011). *C. molmol* Engl. ex Tschirch is used as an effective antimicrobial agent, as an herbal remedy for sore throats, canker sores and gingivitis. Myrrh is also used as a healing tonic, as a stimulant and as a hypolipidemic agent (Urizar et al. 2002; Wu et al. 2002; Ulbright et al. 2005). Traditionally, myrrh is used for the common cold, to relieve nasal congestion and coughing. It is also used for the treatment of wounds and ulcers, especially infections of the mouth, gums and throat (Van Wyk and Wink 2004). It is also applied to abrasions as an antiseptic and anti-inflammatory agent. Myrrh is also found in most Egyptian, Syrian and Roman recipes for drugs and ointments for beauty treatments in antiquity (Ciuffarella 1998). Guggul stimulates the thyroid hormone production, thereby increasing the metabolism of body and causing weight loss (Maheshwari 2010). Further details regarding pharmacological applications of guggul oleo-gum resin and its constituents are summarized in Table 2. Some anti-diabetic and anti-inflamatory activities of specific fractions of guggul have also been patented (Meyer 2000; Bosely et al. 2004).

The exudate of *C. wightii* is a complex mixture of resin (61 %), gum (29.3 %) and other chemicals (6.1 %). More than 150 compounds have been reported and new compounds continue to be reported (Fatope et al. 2003). Guggul comprises of several plant sterols, diterpenes, steroids and alcohols (Verma et al. 1998), however, its main components are E ($C_{21}H_{28}O_2$) and Z ($C_{21}H_{28}O_2$) guggulsterone (El Ashry et al. 2003) which are involved in anticholesterol activity (Satyavati et al. 1969). The two isomers, guggulsterone-E and Z are inter-convertible in callus and cell cultures (Ramawat et al. 2008). Similarly, fungi *Aspergillus niger* van Tieghem and *Cephalosporium aphidicola* Petch., converted guggulsterone-E into guggulsterone-Z and several other compounds (Rahman et al. 1998). Various biochemical studies elucidating the constitution of guggul oleo-gum resin are summarized in Table 3. The components of guggul are known to occur in some other organisms, including some from the aminal kingdom, examples include Z-guggulsterol in *Actinua sulcatus, Cybister tripuncatus* Olivier, *Dytiscus marginalis* L., *Ilybius fenestratus* Fabricius, and the bark of *Khaya grandifolia* Thompson, and guggulsterones in *Ailanthus grandiflora* (Shishodia et al. 2008; Ramawat et al. 2008; Dev 1989).

The hypolipidemic activity of guggul is due to its antagonistic effect against farnesoid X, a nuclear hormone receptor that is activated by bile acids (Urizar and Moore 2003). It is also prescribed to treat rheumatoid and arteriosclerosis (Gujral et al. 1960; Satyavati

Table 2 Pharmacological applications of guggul resin and its constituents

Country	Achievements	Reference(s)
Commiphora molmol Engl. ex Tschirch		
N.E. Kenya	Involved in immunological defence mechanism by reducing lead induced hepatic oxidative damage	Ashry et al. (2009)
Commiphora mukul Engl.		
Canada	Studied the effect of guggulsterone (GS) on human head and neck squamous carcinoma cell lines (SCC4 and HSC2) and found that GS is quite effective (dose and time dependent manner) in inducing apoptosis by releasing bad (inhibitory molecule) from intrinsic mitochondrial pathway. 14-3-3 zeta acts as a target molecule of GS in causing cell death of these chemo-resistant cancerous cells	Macha et al. (2010)
India	Analyzed that guggulsterone is helpful to cure type II diabetes along with its hypoglycaemic effect by improving PPARγ expression in in vivo and in vitro conditions	Sharma et al. (2009)
USA	Reported the efficacy of guggul based Ayurvedic formulations which are commercially marketed as a hypolipidemic drug. After various clinical diagnostic test, it was found that the product of this herbal medicine (more than 80 %) have shown good quality and effectiveness	Yang et al. (2008)
USA	Analyzed efficacy of guggulsterone in maintaince of hypercholesterolemia, antinflammatory and antioxdent activity. Guggul is also effective against different fermenting fungi like *Aspergillus niger* and *Cephalosporium aphidicola*	Deng (2007)
USA	Guggulsterone inhibits tumor cell proliferation, induces S-phase arrest, and promotes apoptosis through activation of c-Jun N-terminal kinase, suppression of Akt pathway, and down regulation of anti-apoptotic gene products	Shishodia et al. (2007)
India	Investigated that guggulsterone induce delipidation by reducing preadipocyte differentiation at low concentration and apoptosis at higher concentration. Also found that cis-GS was more effective at 50 μmol/l concentration than trans-GS in inducing apoptosis. Thus guggulsterone can be helpful for the treatment of obesity by regulating the number and size of fat cells	Singh et al. (2007)
Pennsylvania	Revealed that Z-guggulsterone inhibit the proliferation of human prostate tumor by acting as a potent suppressor and apoptotic agent of angiogenesis in cancerous cells	Xiao and Singh (2007)
India	Revealed that Z-isomer of guggulsterone has more potent cardioprotective and antioxidant activity than E isomer in rats. Also found that Z-guggulsterone improved the synthesis of thyroid hormone and improved free radical scavenging activity	Chander et al. (2003)
Texas	Analyzed that guggulsterone significantly lowered cholesterol-induced hyperlipidemia in rats by reducing the level of triglyceride and phospholipid. Identified that guggulsterone is an active agent against inflammation, obesity and acne. It has exerted stimulatory effects on thyroid metabolism as well as bioavailability of other drugs	Urizar and Moore (2003)
India	Studied the effect of *C. mukul* gum gulggulu in patients of hyperlipidemia with special reference to HDL-cholesterol	Verma and Bordia (1988)
India	Analyzed that guggulipid act as anti-lipidemic agent by reducing cholesterol and lipid levels and strengthen the cardiovascular system	Srivastava et al. (1984)
Commiphora wightii (Arn.) Bhandari		
India	Evaluated that gugulipid in proniosomal gel has enhanced entrapment efficiency which allows the initial faster release of the drug followed by slow sustained release and ultimately possess a long term good anti-inflammatory activity in rats	Goyal et al. (2011)
India	*Vatari guggulu* is effective against lower backache, rheumatism, gout, and sciatica. *Yograj guggulu* is used in rheumatoid arthritis and for removal of toxins from body and for maintenance of healthy metabolism	Pradhan and Dash (2011)
India	Guggulsterone has antispasmodic, expectorant, hypoglycaemic, carminative, anti-suppurative, anti-helmintic, vulneray, livertonic, aphrodisiac, antiseptic, demulcent, depurative, stimulant and diuretic effects. Oil of oleo-gum resin is also used in aromatherapy	Siddiqui (2011)
India	This medicinal herb is used to cure neurological disorders, hypertension and asthma	Maheshwari (2010)

Table 2 continued

Country	Achievements	Reference(s)
India	Demonstrated that guggulipid is a potent inhibitor of LPS induced GFAP expression in C6 cells of rats. Inhibition of this expression is involved in the inactivation of astroglial cells which causes comparatively low production of neuro-inflammatory mediators. Thus revealed the role of this drug in the reduction of inflammation and CNS disorders	Niranjan et al. (2010)
India	Guggulosomes (Ibuprofen loaded) act efficiently as a carrier molecule for drug delivery same as liposomes	Verma et al. (2010)
India, Texas	Analyzed after several clinical trials on a randomized population that guggulsterone is involved in relieving pain and inflammation by inhibiting TNF-induced *COX-2* promoter. It also reduces the proliferation of cancerous cells by suppressing the expression of cyclin D1 protein and promoting the expression of P21 and P27 (cell cycle inhibitor)	Shishodia et al. (2008)
India	Found that guggulsterone prevent the progression of cancerous cells (ovarian cancer cells, lung carcinoma, melanoma, breast carcinoma and head and neck carcinoma) by inhibiting the progression of cell cycle. This is a dose-dependent inhibition process in which expression of cyclin proteins (cyclin D1 and cdc2) is suppressed along with anti-apoptotic genes. DNA synthesis and proliferation was also reduced (80 % reduction within 24 h) in U937 leukemia cells and glee vac-resistant K562 cells respectively	Shishodia et al. (2007)
India	Reported the efficacy of guggul based ayurvedic formulations which are commercially marketed as a hypolipdemic drug. After various clinical diagnostic tests, it was found that the product of this herbal medicine (more than 80 %) has good quality and effectiveness	Singh et al. (2007)
Korea	Revealed that guggulsterone attenuates intestinal pro-inflammatory gene expression by modulating cytokine or LPS-induced NF-kB activation in rat IEC, making it a potential therapeutic bioactive agent for the treatment of tumor and inflammatory bowel disease (IBD)	Cheon et al. (2006)
India	Guggul maintain the cholesterol homeostasis by preventing the transformation of undigested carbohydrate into triglycerides. It enhances the metabolization of existing fats by acting as a burning agent. Thus helpful in preventing Coronary Heart Disease	Jain and Gupta (2006)
	Found that guggul is effective in cancer treatment	Samudio et al. (2005)
Egypt	In vitro studies showed that the fraction of methanolic/ethanolic extract act as a potent inhibitor of Nitric oxide (associated with rheumatoid arthritis and cardiovascular diseases) formation in lipopolysaccharide (LPS)-activated murine macrophages J774 cells	Meselhy (2003)
USA	Analyzed that ethyl acetate extract of *C. wightii* exert cytotoxic activity and antioxidant activity in in vitro condition which was revealed by NMR and GC–MS	Zhu et al. (2001)
India	Estimated that guggulsterone has antioxidative and triiodothyroxin enhancing activity	Panda and Kar (1999)
India	Analyzed that guggulipid can be used to cure acne, respond very effectively against oily faces	Thappa and Dogra (1994)
India	Showed hypolipidemic effects of guggulsterone on rabbits	Satyavati et al.(1969)
India	Prescribed guggulsterone to treat rheumatoidism and arteriosclerosis	Gujral et al. (1960), Satyavati et al. (1969), Malhotra et al. (1970)

et al. 1969; Malhotra et al. 1970). Guggulsterone component has been found to exert antioxidative and triiodothyroxin enhancing activity (Panda and Kar 1999) and is also effective in cancer treatment (Samudio et al. 2005). Medical research on guggul showed that it can be used to cure acne and respond very effectively against oily faces (Thappa and Dogra 1994). This plant has immense potential against various bacterial microbes like *Escherichia coli, Salmonella typhi, Bacillus subtilis, Streptococcus* and different fermenting fungi like *Aspergillus niger* and *Cephalosporium aphidicola* (Deng 2007).

Although guggul is an important and valuable drug for cholesterol lowering, chemical analysis revealed that the insoluble extract (prepared by using petroleum ether, alcohol and ethyl acetate) does not contain any beneficial biological activity rather it may be toxic for animals (Urizar and Moore 2003) and cause some allergic effects like skin rashes and gastrointestinal discomfort (Nohr et al. 2009). Although no clinical data is available for long term safety uses of guggulsterone on human beings, it has been evaluated that there is no mutagenic and acute chronic toxicity in rats, dogs and monkeys rather it was found that negative effects of guggul were associated with high dose (6,000 mg per day) of guggul (Rao et al. 2001). Clinical experiments on some patients showed that the concentration of drugs like the beta blocker propranolol and the calcium channel blocker diltiazem are significantly reduced if they are taken along with guggulipid dose (Shishodia et al. 2008). Aliyu et al. (2007) demonstrated that high doses of ethanolic leaf extract of *C. africana* (A. Rich.) Engl. collected from Northern Nigeria were toxic and caused liver and kidney damage in female albino Wistar.

Problems to be addressed in guggul

Slow growth and over exploitation

Commiphora is a very slow growing genus with small thorny, sturdy, highly branched balsamiferous trees with a short trunk. Oleo-gum resin exudates of *C. wightii* have immense therapeutic values, therefore, excessive and unscientific tapping has led to extensive depletion of the species from nature making the plant vulnerable. Dormant phase is long and the plant remains leafless for long durations. Demand supply gap of guggul oleo-gum resin is increasing very fast. According to an estimate, the demand of guggul is 1,000 MT but the country produces only 100 MT against its requirement (Maheshwari 2010). The oleo-gum resin is collected as it exudates from woody stem of the tree (Haque et al. 2009a). In each collecting season, a guggul tree yields between 250 and 500 g of dry resin (Urizar and Moore 2003) which is extracted from the bark through tapping. A plant generally takes 10 years to reach tapping maturity under the dry climatic conditions. The thick branches are incised during the winter to extract the oleo-gum resin. Mostly the tapping of oleo-gum resin is done by local and tribal people who adopt unscientific methods which generally results in the death of plants. During harvesting they apply a paste around the incision consisting of horse or wild ass urine, oleo-gum resin and copper sulphate. Whilst this crude method increases the amount of oleo-gum resin three to four times over that obtained under normal tapping procedures, the shrub subsequently becomes unfit for tapping for the next couple of years and ultimately the plants may die due to the injurious effect of copper sulphate (Kshetrapal and Sharma 1993). Patel et al. (2008) emphasized the necessity to tap the plant at the right time in a year to obtain maximum oleo-gum resin production. Moreover, due to the presence of oleo-gum resin, the wet stem burns very well so it is also used as fuel wood (Haque et al. 2009a). Recently Siddiqui (2011) reported that guggul production can be enhanced by up to 22 times with an improved tapping technique using "Mitchie Golledge knife" coupled with ethephon (2-chloroethyl phosphonic acid), a plant growth regulator. However, excessive production of guggul through ethephon application exhausts the plant and kills it in the long term. Onset of summer, April and May, were established as the best period for guggul production being a stress induced secondary metabolite based on epifluorescence microscopy (Bhatt et al. 1989). However, ethrel did not increase the guggulsterone content in cell cultures of *C. wightii*, indicating a senescence effect on the in vitro plant (Shishodia et al. 2008). According to Siddiqui (2011) a healthy guggul tree can yield about 700–900 gm oleo-gum resin.

Due to lack of cultivation, natural regeneration is almost negligible as compared to its depletion in nature. Attention is required to prevent its exploitation and for sound management and conservation. This will ensure survival of this important oleo-gum resin species heading towards extinction and will also save the fragile arid and semi-arid lowlands from desertification and other environmental vulnerabilities (Lemenih and Teketay 2003). Being a very slow growing plant, returns from the plant are only after 7–15 years and thus is not preferred as a social forestry tree species.

Pests and pathogens

It is estimated that around 10 % seedlings of *C. wightii* at Mangliawas, National Herbal Farm in India are destroyed by insects and pests (Chaturvedi et al.

Table 3 Biochemical studies elucidating the chemical constitution of guggul resin

Species	Area of collection	Achievements	Reference(s)
C. africana (A. Rich.) Engl.	–	Identified a dihydroflavonolic compound glucoside phellamurin in the fraction of crude extract	Ma et al. (2005)
C. africana	Sub Saharan Africa, Northern Nigeria	Identified phenolic compounds and tannins in ethanolic leaf extract while hexane fraction constituted alkaloids, steroids and triterpenes	Aliyu et al. (2002)
C. africana	Benin	Used GC, GC–MS and 13C-NMR spectroscopy to identify that essential oil contained high amount of sesquiterpenes, among which α-oxobisabolene (61.6 %) and γ-bisabolene (10.0 %) are two most important and major sesquiterpenes	Ayedoun et al. (1997, 1998)
C. angolensis Engl.	–	Revealed that the characteristic colour and precipitation reactions are due to the presence of condensed tannins in the alcoholic extract of the bark	Cardoso do Vale (1962)
C. confusa Vollesen	Kenya	Analyzed the presence of two new dammarane triterpenes in the resin	Dekebo et al. (2002)
C. cyclophylla Chiov. ex Guid.	Southern Ethiopia	Identified monoterpenoid hydrocarbons in which limonene is the major component; resin is also rich in sesquiterpenes	Abegaz et al. (1989)
C. dalzielli Hutch.	Ghana	Identified seven dammarane triterpenes from stem bark extract prepared in petroleum	Waterman and Ampofo (1985)
C. erlangeriana Engl.	Ethiopia, Somalia	Revealed the chromatogram of resinous powdery extract (prepared in a mixture of methanol–ethyl acetate) and identified four new lignans, erlangerins (A to D). These ligans are cytotoxic to mammalian cells and traditionally are used as arrow poison	Habtemariam (2003), Dekebo et al. (2002)
C. guidotti Chiov. ex Guid.	–	Identified seven sesquiterpene and one furanosesquiterpenoid from essential oil of isolated and HPLC purified ethyl acetate extract. Structural configuration of three main compounds α-santalene, α-bisabolene and furanodiene was elucidated by MS, 1H-NMR and 13C-NMR technique. Pharmacologically active (+)-T-cadinol (sesquiterpene) was found to be involved in smooth muscle relaxation	Claeson et al. (1991), Craveiro et al. (1983)
C. guillaumini H. Perrier	Medagascar	Identified an important compound 1,2-dioleoylglycerol in arils of the seeds which is a potent ant attractant	Schmeer et al. (1996)
C. holtziana Engl.	Kenya	Identified one novel furanogermacrene compound from the resinous exudate	Cavanagh et al. (1993)
C. incisa Chiov.	Kenya	Characterized two aryltetralin lignans and four triterpene derivatives (in oleo-gum resin) by chromatographic analysis of diethyl ether extract. Also found two novel triterpenes which were identified as 1α-acetoxy-9,19-cyclolanost-24-en-3β-ol and 29-norlanost-8,24-dien-1α,2α,3β-triol	Provan and Waterman (1988)
C. kataf Engl.	Kenya	Identified camphene (0.8 %), g-elemene (2.4 %), b-elemene (14.4 %), germacrene-D (15.5 %), elmol (4.6 %), furanogermacra-1, 10 (15)-diene-6-ones (16.7 %) in steam distillation of essential oil using GC–MS and NMR	Baser et al. (2003)
C. kua (R.Br. ex Royle) Vollesen	Kenya, Ethiopia, Somalia	Four active compounds—mansumbinone, mansumbinoic acid, picropolygamain and lignan-1 containing anti-inflammatory activity were identified from the extract. These are also involved in the prevention of myeloperoxidase formation	Battu et al. (1999)

Table 3 continued

Species	Area of collection	Achievements	Reference(s)
C. kua	Kenya, Ethiopia, Somalia	Steam distillation of volatile oil contains many compounds viz: α-thujene, β-pinene, p-cymene, limonene, sabinene, car-3-ene, myrcene and terpinene-4-ol. Also identified many known furanosesquiterpenoids by chromatographic analysis of ethyl acetate extract	Manguro et al. (1996)
C. kua	Kenya, Ethiopia, Somalia	Identified three triterpenes viz: mansumbin-13(17)-en-3, 16-dione, 3β-hydroxy mansumbin-13 (17)-en-16-one and 16 oxo-mansumbin-3(28), 13(17)-dien-3- oic acid from petroleum extract of stem bark	Provan et al. (1992)
C. merkeri Engl.	–	Identified a new pentacyclic triterpene, which showed a potent anti-inflammatory and analgesic activity	Fourie and Snyckers (1989)
C. mukul Engl.	–	Estimated three new polypeptide type triterpenes viz: myrrhanol B, myrrhanones B, and acetate A in the methanolic extract of oleo-gum resin. Also found a new octanordammarane type triterpene, epimansumbinol along with other 17 known compounds. Revealed a rapid and potent anti-inflammatory activity of 50 % aqueous methanolic extract in mice granulomas	Matsuda et al. (2004)
C. mukul	–	Identified 14 new compounds from methanolic extract of oleo-gum resin, of which seven are: IE, 4E, 8E-tritene; 2E, 12E-dien-7-ol; 4Z, 6E-diene-12-ol; 16-dione; 13E, 17E, 21E-tritene-3-one; 13E, 17E, 21E−18-diol; 1, 3-benzodioxole are hexane soluble. In contrast, the resin is completely devoid of triterpenoids	Francis et al. (2004)
C. mukul	–	Isolated the steroidal nucleus of the active ingredient of guggulsterone in ethyl acetate extract	Sarkhel et al. (2001)
C. mukul	USA	Analyzed ketonic fraction of ethyl acetate extract containing E and Z guggulsterone	Deng (2007)
C. myrrha Engl.	Ethiopia	Identified curzerene, curzerenone, furanoeudesma- 1, 3- diene1,10 (15)-furanodien-6 one	Lemenih and Teketay (2003)
C. myrrha	–	Characterized Myrrhanol A, and myrrhanone A on the basis of physiochemical analysis	Kimura et al. (2001)
C. myrrha	–	Evaluated that the resinous gum is composed of herrabomyyhols, heerboresene, commiferin, kertosteroids, compesterol, sitosterol commiphoric acid and commiphorinic acid	Rao et al. (2001)
C. myrrha	–	Evaluated that hydrolysis of water soluble gum is composed of mainly arabinose, galactose, xylose sugars and mythylglucuronic acid	Evans (1989), Leung (1980)
C. ornifolia (Balf. f.) J. B. Gillett	Soqotra Island in Yemen	Identified the chemical composition of oil by GC/MS and found that the oil contains high amount of oxygenated monoterpenes (56.3 %), camphor (27.3 %), α-fenchol (15.5 %), fenchone (4.4 %) and borneol (2.9 %)	Mothana et al. (2010)
C. parvifolia Engl.	Soqotra Island in Yemen	Identified the chemical composition of oil by GC/MS and found that the oil contains mainly oxygenated sesquiterpenes (36.1 %) and aliphatic acids (22.8 %) which consist of caryophyllene oxide (14.2 % main constitute), β- eudesmol (7.7 %), bulnesol (5.7 %), T-cadinol (3.7 %) and hexadecanoic acid (18.4 %)	Mothana et al. (2010)
C. pyracanthoides Engl.	Southern Africa	Analyzed that ether extract of oleo-resin gum is source of triterpene acids, available in two free forms viz: Comic acid A, B, C, D and E and combined form like glycosides	Thomas (1961)

Table 3 continued

Species	Area of collection	Achievements	Reference(s)
C. rostrata Engl.	Kenya, Somalia, Ethiopia	Characterized the essential oil of stem bark by gas chromatography and found that it contain 30 bioactive components which have a major role in elimination of pests and pathogens. The activity of these aliphatic ketones and aldehyde is more, than the commercially marketed insect repellent DEET	McDowell et al. (1988)
C. tenuis Vollesen	Ethiopia	Revealed the presence of four free triterpenes in the extract using column chromatography. Also found another 37 compounds in the essential oil of oleo-resin	Asres et al. (1998)
C. terebinthina Vollesen	Northern Kenya	Identified monoterpenoid hydrocarbons in which limonene is the major component; also rich in sesquiterpenes	Abegaz et al. (1989)
C. wightii (Arn.) Bhandari	USA	Identified 17 different metabolites from commercial guggul samples including three new compounds—20(S),21-epoxy-3-oxocholest-4-ene,8 β-hydroxy-3,20-dioxopregn-4,6-diene and 5-(13$'$ Z-nonadecenyl)	Ahmed et al. (2011)
C. wightii	India	Estimated the presence of essential oils (0.37 %), myrecene, dimycerene and polymyrecene compounds by chromatographic analysis	Jain and Gupta (2006)
C. wightii	–	Isolated a new antifungal flavone compound, muscanone (along with naringenin) from ethanolic extract of air-dried trunk using silica gel chromatographic analysis. Found that muscanone has anti-microbial activity against *Candida albicans*, confirmed by MTT assay	Fatope et al. (2003)
C. wightii	Egypt	Identified eight pharmaceutically applicable compounds from ethanolic extract of oleo-gum resin viz : a) Z-guggulsterone, E-guggulsterone, guggulsterone-1, myrrhanol-A and myrrhanone-A were previously known while b) Guggulsterone-M, dihydro guggulsterone-M and guggulsterol-Y are new components. Structure was analyzed by spectroscopic method	Meselhy (2003)
C. wightii	–	Characterized two ferrulates (having in-vitro cytotoxicity) by column chromatography in ethyl acetate extract, and elucidated that one ferrulate guggulsterone is maintained in 2S, 3S, 4R configuration. Alcoholic hydrolysis of these ferrulates were found to be a mixture of (Z)-5-tricosene-1, 2, 3, 4-tetrol and (Z)-5-tricosene-1, 2, 3, 4-tetraol which exhibit antioxidant activity	Zhu et al. (2001)
C. wightii	India	Identified several plant sterols, diterpenes, steroids and alcohols	Verma et al. (1998)
C. wightii	India	Identified guggulsterone IV	Purushothaman and Chandrasekharan (1976)
C. wightii	India	Identified diterpene alcohol (melting point—37–38 °C)	Prasad and Sukh Dev (1976)
C. wightii	India	Identified guggulsterone I, II, III, V, Z, E and ctadecan-1,2,3,4-tetrol	Patil et al. (1972, 1973)

1987). Whiteflies, *Bemisia* sp., harm guggul plant up to some extent by feeding on sap of leaves (Dalal 1989). Leaf-eating caterpillar also harms the leaves of this plant. *C.wightii* plants are also prone to damage by termites in the drier months preceding winter and after tapping of oleo-gum resin. In termite infestation a hole is made through the buried end of the stem and tunnel is bored into cuttings or root of the plants. Termite infestation causes death of the plants after yellowing and wilting of leaves. For elimination of termite, a mixture of 5 % elderine with water is most effective although kerosene or carbon disulphide is also beneficial. Powder of neem leaves mixed with soil and gammexane (Benzene hexachloride) powder can also inhibit the attack of termites. Mercuric chloride (0.25 %) or Copper sulphate (0.55 %) in aqueous solution and Haptafan (3 %) also has potent effect against termites (Bhatt and Dixit 1974).

Fungal leaf spot disease caused by fungus *Phoma glomerata* (Corda) Wollenw. et Hochapfel is a serious problem for *Commiphora* are perhaps is the most prevalent plant disease in Northeastern India. Initial stage of the disease start with the development of small black concentric ring-spots on leaf surfaces followed by brown discoloration of tissues, ultimately turning into black concentric rings (Sharma and Gour 1987). As this plant requires arid and semi-arid land with rocky tracts, overwatering especially during rainy season can cause root-rot disease due to lack of gaseous exchange and causing appearance of brown lesions on leaves. These leaves further turn yellow, wilt and finally the plant dies due to arrested growth.

Dimorphic plants

The species is dimorphic with male, female and polygamous plants and the flowers may be uni- or bisexual, with the unisexual flowers only being semi-developed with non-functional stamens (Rao et al. 2001). Gender in guggul seedlings can be determined only during flowering. Large number of isolated groups of female individuals, only one andromonoecious and two exclusively male plants were recorded by Gupta et al. (1996) during a field examination in India. It was also revealed that female plants could set seed irrespective of the presence or absence of pollen. Hand-pollination experiments and embryological studies confirmed the occurrence of non-pseudogamous apomixis, nucellar polyembryony and autonomous endosperm formation for the first time in this plant (Gupta et al. 1996). They documented the formation of seeded fruits from female flowers with no access to pollen or even after emasculation and bagging; failure of pollen tubes to enter the ovule following hand pollination and autonomous development of endosperm. Gupta et al. (1998) described pollen-stigma interaction as the basis of low seed set in *C. wightii*. They reported that the pistil did not support pollen tube growth, perhaps due to alteration in the orientation of cells of the transmitting tissue and absence of proteins in the intracellular matrix resulting in poor seed set. Prakash et al. (2000) reported multiple sapling formation by germination of polyembryonic seeds of *C. wightii*. Seed set is approximately 16 % in the plants in the Aravalli ranges and even lower in drier parts of Western Rajasthan (Kumar et al. 2003). Production of seeds through apomixis bypassing the process of fertilization will produce plants which have a genetically identical makeup. This process ensures multiplication of the plant in nature however; such a population will be highly vulnerable to wipe out if a disease epidemic occurs in the region. Moreover, the breeding programs cannot be undertaken in the absence of genetic variability. Though apomixis occurs in *C. wightii*, Gupta et al. (1996) reported that apomixis is not the main method of reproduction based on the availability of variability in field grown plants. The seeds are produced by cross pollination resulting in hererozygous population. Therefore, study of genetic variability using biochemical and molecular methods is must to pin point superior germplasm which can be used for cloning to produce high yielding planting population which can be later used in social forestry and reforestation programs. Moreover, identification of male plants is must for breeding purposes.

Conventional propagation methods

Commiphora wightii primarily grows well in rocky tracts of arid and semi-arid lands along with some coastal areas of loamy soil with pH ranging from 4 to 8 (Kant et al. 2010). Traditional propagation methods include seeds and stem cuttings. In nature, regeneration of *C. wightii* through seeds is very poor. The plant is dimorphic and number of male plants in nature is very limited. Gupta et al. (1996) reported the presence of only two male plants along with a large population of female plants, thus limiting the availability of pollens for fertilization. Seed maturation depends upon the development of cellular endosperm. The germination of seed is hampered by the hard endocarp which does not allow entry of water, gaseous exchange and embryo expansion. This harsh natural condition is responsible for poor seed germination rate of only 1.4 % (Yadav et al. 1999). Therefore, mechanical scarification of seed coat is beneficial to improve the germination rate. Seeds are sown in the months of June and July for germination in monsoon rains. After one month, these seedlings are transferred into poly bags, and after 2 years the germinated seedling attain a height of 30–50 cm and are ready for field transfer. In India, Guggulu Herbal Farm at Mangliawas, Ajmer, Rajasthan is the only National Herbal Farm, where thousands of guggul plants are growing. In spite of this, it is difficult to find plants which have been developed through seeds. Propagation through seeds is more advantageous because seedlings have a deeper root system compared to the plants established through cuttings (Yadav et al. 1999). Poor germination rates coupled with high mortality rate of seedlings limits propagation of *Commiphora* by this method. Fruit set and yield of fruits per plant are very low in natural conditions. Poor seed set, very poor seed germination (5 %) and harsh arid conditions are responsible for complete failure of plant establishment from seed in nature. Natural regeneration through seeds has been rarely observed below the parent plants in the farm and in the forests (Yadav et al. 1999).

Due to global warming, temperature is increasing and rainfall is decreasing. The natural habitat of Guggul is dry/arid or highly salty where bacterial growth is normally slower as compared to other areas. This directly affects the microflora of soil, which is usually helpful in damaging the endocarp and increasing the permeability of endocarp to water. Due to prevailing drought conditions soil moisture has gone down below the critical level, inadequate to support seed germination under natural conditions.

Propagation of *C. wightii* is usually done through cuttings. The sprouting and establishment of cuttings varies between 60 and 100 %. The cuttings are usually planted during late summer from healthy and disease free branches, when the plant is almost leafless. With the onset of monsoon, when foliation growth/flowering starts, the plant becomes physiologically active and the cuttings also show signs of sprouting in 25–50 days. Treatment with root hormones, viz., Credik-1 and Credik-2 in July and August, was found beneficial for development of roots. After establishment in nursery beds the plants are transferred to poly bags and then planted in the field during rainy season in the second -third week of June (Yadav et al. 1999).

Air layering also known as "Gootee layering or Marcottage-a layering" is being practiced by Indian and Chinese people since ancient periods (Fig. 1d). Adaptation of this technique is increasing due to limitations in the above mentioned propagation methods (Yadav et al. 1999).

Efforts have been made by government organizations like Department of Environment, Council of Scientific and Industrial Research (CSIR), State forest departments and Ministry of health and family welfare in particular to conserve this species. Establishment of Guggulu Herbal Farm, Mangliawas, Ajmer, Rajasthan is an important step towards extensive cultivation of *C. wightii*. These efforts though substantial are not sufficient in view of the increasing demand of the raw drugs and to cope up with the fast denudation of the forests and depletion of the species in nature. Further efforts need to be made to conserve this species in nature and also to propagate it extensively. Further, forest areas with more abundance of guggulu deserve urgent attention to conserve them in nature as "Biosphere Reserves" before they are destroyed by over exploitation (Yadav et al. 1999).

Breeding

Commiphora wightii is mostly found wild. Efforts have recently been initiated to cultivate this plant by Government of India. Guggulu Herbal Farm, Mangliawas is the only example of guggul cultivation. However, most of the plants therein have been raised through cuttings from female plants. Therefore, the germplasm needs to be characterized before any breeding program can be initiated for guggul. Since guggul is dioecious, identification of superior paternal parents is a pre-requisite for achieving the breeding goals. However, selecting parents is still a major problem. The conventional breeding programs for improving quantitative traits require large inputs of labor, land and financial resources. Therefore, plant breeders must identify promising lines as early as possible in the selection process. Marker assisted selection (MAS) is an excellent tool for selecting beneficial genetic traits in crops and forest trees at an early stage as well as for assessing the genetic potential of specific genotypes prior to phenotypic evaluation (Kalia et al. 2011). However, non-availability of mapping populations and substantial time needed to develop such populations may prove to be a major limitation in the identification of molecular markers for specific traits. Additional limitations to this technology include limited availability of gene pool, restricted distribution of species and lack of

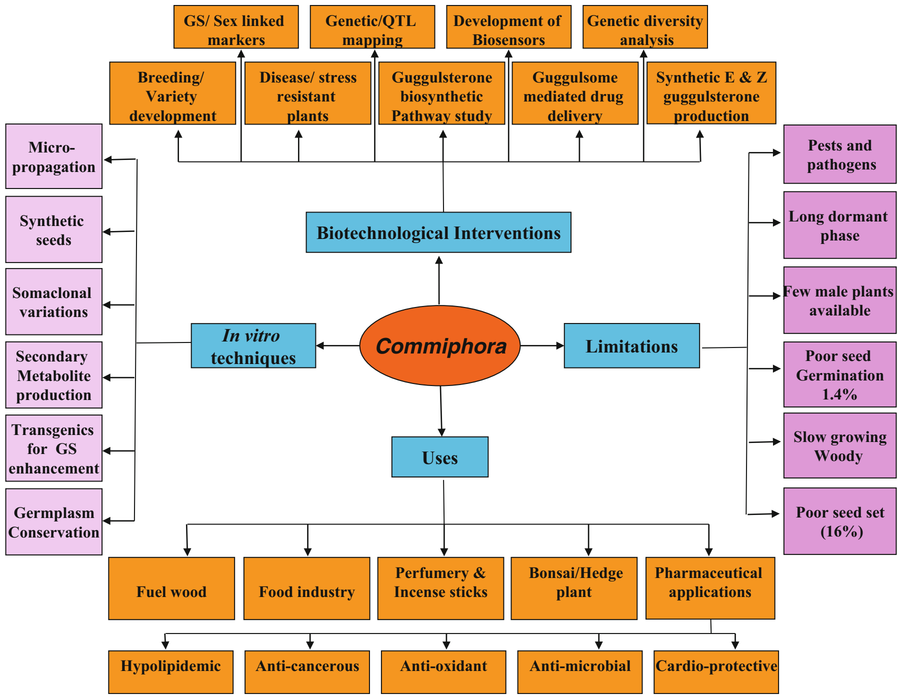

Fig. 2 Limitations, uses and tentative applications of in vitro techniques and biotechnological interventions in *Commiphora*

knowledge about extent of hybridization possible between species, non-availability of linkage maps and lack of markers associated with important QTLs for guggulsterone production.

Progress made using modern biotechnological tools

Due to the commercial importance, extensive use in medicine, faulty and unscientific techniques of tapping employed by local inhabitants, over-exploitation for oleo-gum resin, slow growth and lack of organized or systematic cultivation, the wild population has declined fast and this species has been listed as threatened. The conventional methods of propagation using seeds and vegetative cuttings are in place but are very slow. Moreover, such methods are not suitable for large scale multiplication as stock material with sufficient biomass is not available. Also, survival percentage of cuttings is variable and is affected by seasons. Therefore, there is an urgent need to conserve this species *ex situ* through in vitro method and to develop reliable and rapid protocol for its micropropagation. In spite of being pharmacologically important, *Commiphora* had not gained much attention till approval of its use in food products by US-FDA in 1994. Recently, extensive studies have been reported on pharmaceutical applications as well as regarding biochemical and molecular characterization of guggul. However, application of modern biotechnological tools for improvement and multiplication of this important oleo-gum resin tree heading towards extinction are still in a stage of infancy. An attempt has been made to summarize the available literature on *C. wightii* and the attempts being made for improvement of this important tree. Figure 2 summarizes the limitations, uses and the possible areas wherein biotechnological interventions can be made for improvement of *Commiphora*.

In vitro culture technology (micropropagation)

Micropropagation or in vitro culture is a biotechnological approach for vegetative propagation of elite genotypes for mass multiplication and genetic improvement programmes, especially in dioecious

Table 4 Application of in vitro techniques in *Commiphora wightii* (Arn.) Bhandari

Explants	Response	Reference(s)
Micropropagation		
Cotyledonary nodes, Microshoots	Rooted plantlets produced	Kant et al. (2010)
Nodal segments	Rooted plantlets produced	Soni (2010a)
Shoot tips, nodes, internodes and leaves	Rooted plantlets produced	Singh et al. (2010)
Nodal segments	Shoot formation	Barve and Mehta (1993)
Cotyledonary nodes	Shoot multiplication	Yusuf et al. (1999)
Somatic embryogenesis		
Immature fruits	Somatic embryo proliferation	Kumar et al. (2006)
Callus culture	Somatic embryos	Kumar et al. (2003)
Immature zygotic embryos	Somatic embryos	Singh et al. (1997)
Immature fruits	Torpedo stage somatic embryos	Kumar et al. (2004)
Secondary metabolite production		
Cell suspension culture	Guggulsterone accumulation in vitro	Suthar and Ramawat (2010)
Zygotic embryos	Calcium deprivation markedly enhances guggulsterone accumulation in cell cultures	Dass and Ramawat (2009a)
Zygotic embryos	Elicitation of guggulsterone production in cell cultures by plant gums	Dass and Ramawat (2009b)
Zygotic embryos	Studied somatic cell variability for guggulsterone production in cell suspension culture	Dass and Ramawat (2009c)
Zygotic embryos	Anthocyanin production in callus cultures	Dass et al. (2008)
Callus cultures	Morphactin influences the production of guggulsterone	Tanwar et al. (2007)
Zygotic embryos, leaves and stem	Guggulsterone production in callus cultures	Mathur et al. (2007a)
Suspension cultures	Guggulsterone production in shake flasks and bioreactors	Mathur et al. (2007b)
Immature fruits, somatic embryos	Guggulsterone production in cultures	Kumar et al. (2006)
Immature fruits	Development of resin canals in callus cultures and torpedo stage somatic embryos	Kumar et al. (2004)
Leaf	Production of guggulsterone in immobilized cells	Phale et al. (1989)
Leaf	Production of guggulsterone in callus cultures	Barve and Mehta (1987)

species like *C. wightii*. It can be used as an alternative technique for conservation of this endangered plant species as well as for production of highly valuable compound guggulsterone in vitro. Moreover, the modern biotechnological tools can be exploited to improve the content and quality of guggulsterone, only when a protocol for regeneration of complete plantlets in vitro is available (Kalia et al. 2011). Two distinct patterns of in vitro differentiation, i.e., organogenesis and somatic embryogenesis have been used for micropropagation of *C. wightii* (Table 4).

Organogenesis

Little information is available on clonal propagation of *C.wightii* because it is highly difficult and recalcitrant species. The success of culture is largely governed by the quality of explants, genotype, physiological state of the tissue, and the time of the year when the explants are collected. Seasonal changes have a great impact on bud spourting, explant contamination and establishment (Hu and Wang 1983). *C. wightii* is quite recalcitrant towards in vitro conditions as has been depicted by the results of research programs undertaken till now. Nothing concrete has come out from different programs on micropropagation of *C. wightii* since 1979 (Kumar et al. 2003; Sharma et al. 1998). Sterilization of explants collected from the field grown plants remains a major limitation in establishment of *C. wightii* in vitro due to the presence of oleo-gum resin on explant (stem, leaf and petiole) and bacteria in the resin canals (Ramawat et al. 2008). The plant has a long dormant phase so availability of tender shoots for establishment remains a limitation. Phenolic exudation further limits the establishment of explants in vitro. Moreover, no selected germplasm is available for *C. wightii* as it is a wild plant. Zygotic embryos are preferred for in vitro establishment of *C. wightii*, however, low seed set, difficulty in dissecting hard fruit to obtain ovule, and low frequency of response of zygotic embryos makes it a difficult material (Kumar et al. 2003). Organogenesis has been induced through axillary shoot proliferation from nodal segments (Barve and Mehta 1993; Soni 2010a), seedling explants (Yusuf et al. 1999; Kant et al. 2010) and shoot tips, nodes, internodes and leaves (Singh et al. 2010).

In *C. wightii,* media with different concentrations and combinations of growth hormones were tried for the establishment of nodal as well as shoot tip explants. Highest frequency of shoot formation was achieved on MS medium supplemented with BAP (17.8 μM), Kinetin (18.6 μM) glutamine (100 mg l^{-1}), thiamine HCl and 0.3 % activated charcoal. However, for multiple shoot bud formation modified MS medium supplemented with lower concentration of Kinetin and BAP was used. The rate of successful multiplication is quite low when the explants are derived from mature slow growing woody shrub (Barve and Mehta 1993). Multiple shoots were developed from nodal segments on modified MS medium with two cytokinins, BAP (3.0 mg l^{-1}), Kinetin (0.5 mg l^{-1}) along with IBA (0.5 mg l^{-1}) (Soni 2010a). Barve and Mehta (1993) achieved rooting of shoots on medium containing half strength MS salts and supplemented with different auxins IAA, IBA and NAA added singly and in combinations. They reported that long term exposure to higher auxin levels can cause inhibition of further root growth. So it was essential to transfer the explants to complete darkness in auxin free medium for promoting higher frequency of rooting. However, Kant et al. (2010) reported efficient rooting on White's medium without any hormones and high concentration of charcoal. Acclimatization or hardening was an important factor for evaluating the survival rate of tissue culture rasied plants. Hardening was done in mist chamber with 85–95 % relative humidity. Soil with farm yard manure in the ratio of 1:1 proved beneficial for establishing the plantlets in field (Kant et al. 2010). Soni (2010a) used soil and manure in 1:3 ratio while Barve and Mehta (1993) transferred the plantlets in sterile vermiculite and sand (1:1) with one quarter strength MS salts. Only 5 % survival rate of acclimatized plantlets (5–6 week-old) was reported when they were transferred to soil and grown in the nursery (Soni 2010a).

Limited number of plantlet production has been achieved in all the above studies and also their rate of establishment in the field has been low. These studies have demonstrated the feasibility of micropropagation technology in *C. wightii,* however; more efforts are required to develop a protocol for mass multiplication so that plants may be produced for large scale plantations. Therefore, further systematic scientific studies are necessary to focus on improving the multiplication protocol and survival rates of the plantlets in the greenhouse to evaluate its efficiency on commercial scale. In spite of many reports on micropropagation of *Commiphora wightii*, field performance of tissue culture raised plants has not yet been initiated.

Somatic embryogenesis

Somatic embryogenesis can potentially produce large number of plantlets in a short time therefore, has always been preferred over vegetative propagation by organogenesis and axillary bud proliferation. Moreover, root and shoot meristems are produced in a single step unlike organogenesis where rooting follows shoot production. In addition, the process of somatic embryogenesis can be scaled up in a bioreactor using cell cultures, and development of artificial seeds through immobilization (Jain et al. 2002). Kumar et al. (2003) achieved somatic embryogenesis by repetitive reciprocal transfer of callus cultures of *C. wightii* between basal medium and MS medium containing 2,4,5-trichlorophenoxy acetic acid (2,4,5 T) and kinetin. They found that immature zygotic embryos were the only suitable explants for somatic embryogenesis. The somatic embryos germinated into plantlets which were successfully transferred to field conditions. Kumar et al. (2006) reported secondary somatic embryogenesis in *C. wightii* on basal medium. Asynchronous embryo development, low conversion rate of embryos into plantlets and survival of plantlets remains the major limitations of somatic embryogenesis and need to be resolved before this technique can be used for mass multiplication of *C. wightii*. Once standardized, somatic embryogenesis can be used for synthetic seed production and for secondary metabolite production also. In addition somatic embryos can be employed for conservation and cryopreservation of *C. wightii* also.

Genetic transformation studies have not yet been initiated in *Commiphora wightii* due to lack of standardized micropropagation protocols and field trials. Somaclonal variations which may appear during regeneration of plants from callus cultures can be exploited as a potential tool for increasing variability thereby widening the genetic base for breeding programs.

Secondary metabolite production

Rutheless exploitation of *C. wightii* has made it endangered and the natural resources are highly limited and insufficient to meet the current demand of guggul oleo-gum resin which is the material of commerce and contains only 2–3.3 % guggulsterone (Ramawat et al. 2008). Therefore, alternative biotechnological tools for production of guggulsterone need to be investigated. Moreover, from among 185 species of *Commiphora, C. wightii* is the only species which produces guggulsterone, therefore it becomes essential to develop a high throughput method for production of guggulsterone in vitro (Mathur et al. 2007b; Barve and Mehta 1993). Tissue culture is the only available method for production of secondary metabolites via cell, callus culture and cloning of selected high yielding guggul varieties. However, cytodifferentiation in callus and cell cultures is a prerequisite for the production of secondary metabolites, which are produced in complex tissue systems like laticifers and resin canals (Ramawat et al. 2008; Suri and Ramawat 1996). Resin canal formation has not been reported in callus cultures, but the same was observed during somatic embryogenesis. Kumar et al. (2004) reported the formation of resin canals in torpedo shaped and cotyledonary stage somatic embryos of *C. wightii*. They reported that the early stage somatic embryos were devoid of resin canals. The resin canals formed in the somatic embryos were comparable to those produced in the stem and these somatic embryos accumulated three times higher level of guggulsterones. Kumar et al. (2006) reported that the in vitro produced aseptic resin canals of somatic embryos can be used for guggulsterone production alleviating the pressure on natural resources. Tanwar et al. (2007) also reported that undifferentiated callus cultures contained lowest amount of guggulsterone compared to embryogenic cultures and plant parts.

Although, Kumar et al. (2004 and 2006) reported the formation of resin canals in torpedo stage somatic embryos, which accumulated high amount of guggulsterone but due to slow growth of these cultures prime objective of production of high guggulsterone content was not fulfilled. For this purpose different researchers used various growth regulators and growth retardants which act synergistically to enhance of guggulsterone production under in vitro conditions (Tanwar et al. 2007). Zygotic embryo and leaf explant derived callus cultures have been used for optimization of guggulsterone production in vitro (Table 4). Mathur et al. (2007a) explored the possibility of guggulsterone production in cell cultures without cytodifferentiation. Maximum callus culture was obtained after 35-days growth on modified MS medium. Maximum guggulsterone production was recorded in the tissues grown on medium containing 340 mg l^{-1} KH_2PO_4, 110 mg l^{-1} $CaCl_2$, 40 g l^{-1} maltose (60 μg/g dry weight) followed by sucrose-glucose combination (45 μg/g dry weight) (Mathur et al. 2007a). They found that guggulsterone production was enhanced (threefold) after addition of sugars (either sucrose or maltose @ 40 g l^{-1}). Thus the nutrients like nitrogen, phosphate, calcium and sugars have profound effect on in vitro production of guggulsterone. Variation in colour, texture, growth and guggulsterone content was recorded among different cell lines grown on the same medium (Dass et al. 2008). Callus varied from white to brown and pink to dark red in colour. White, soft and watery cultures contained least guggulsterone levels. However, no correlation was found between red colour (anthocyanin production) and guggulsterone content. Growth and guggulsterone content was almost stable in majority of the clones in every passage over a span of one and half years. Decrease in growth was associated with increase in guggulsterone-E and Z, as well as total guggulsterones accumulation (Dass et al. 2008). This study demonstrated the promising role of cell and tissue culture in production of guggulsterone in vitro.

Ramawat and Mathur (2007) showed the inter-relationship between primary metabolism and secondary metabolite formation. Reduced growth (by growth retardants) stimulates the photo-assimilation power of the biosynthetic pathways involved in secondary metabolite production. Guggulsterone production was enhanced (fivefold) in the presence of fungal elicitor along with the stimulatory effects of growth retardants chloramequet chloride (CCC) and N, N-dimethylaminosuccinamic acid (ALAR) after 5th day and 10th day of inoculation respectively, while by 17th day of culture, the concentration of guggulsterone could be increased up to 353 mg l^{-1} in a two-stage batch-fed process (Suthar et al. 2008). Growth retardant CCC prevent the cyclization of geranylgeraniol, leading to the inhibitory effect on gibberellin biosynthesis, while ALAR inhibits the growth of plant as well as its biomass (Suthar and Ramawat 2010; Barsa 1994).Various plant growth retardants viz: CCC, ethrel and triacontanol play an important role in regulating the growth of plant and secondary metabolite production such as essential oil components ajmalicine, vindoline, artemisinin, serpentine and tabersonine in aromatic and medicinal plants (Haque et al. 2007; Sangwan et al. 2001). Guggulsterone production in callus culture was enhanced in the presence of morphactin (Tanwar et al. 2007). Maximum callus growth was reported on medium containing a combination of morphactin and 2iP (0.1 and 2.5 mg 1^{-1} respectively), while increasing guggulsterone production, mainly Z-form (eight folds) was obtained by using morphactin and 2iP, at lower concentration (0.1 and 1.0 mg 1^{-1} respectively) in non-embryogenic callus cultures. Concentration of calcium ions also had an important role in accumulation of guggulsterone in cell cultures. If ionic concentration of Ca^{2+} was decreased to 0.5 mM, production of guggulsterone would be enhanced up to 66 % (Dass and Ramawat 2009a). Immobilized cells and callus can also be used for in vitro production of guggulsterone E and Z (Kumar et al. 2006).

Production of oleo-gum resin is a stress induced phenomenon. Both biotic and abiotic stresses induced by various pathogens or elicitors viz: methyl jasmonate, ethrel and salicyclic acid was observed to enhance the production of secondary metabolite guggulsterone in intact plants as well as in tissue culture. Dass and Ramawat (2009b) reported for the first time that the gum (a complex mixture of polysaccharides and arbinogalactan) can act as an elicitor in increasing the amount of guggulsterone. Mesquite gum enhanced the guggulsterone production (2–3 fold) at the concentration of 50 mg l^{-1} while Gum Arabic increased the content (1.5–2.5 fold) at the concentration of 500 mg l^{-1}. Higher concentration had toxic effects on guggulsterone production. It was a time dependent study in which guggulsterone content increased during 6–9th day of culturing which declined after 12th day. This study provided a valuable alternative for in vitro production of highest guggulsterone (maximum amount 285 mg l^{-1} in 3 l flasks) in a very short duration of 11 days (Dass and Ramawat 2009a).

Mathur et al. (2007a) observed a correlation between yield of guggulsterone in callus culture and in vivo in the plant and revealed after quantification by molar extinction coefficient method that the amount of guggulsterone was six times lower in cultured cells than in intact stem. The content of guggulsterone of callus was maximum during January to July, the period of gum exudation in nature.

As yet, undifferentiated cell suspension cultures have only been able to produce 200 μg/l guggulsterone in a 2 l stirred-tank bioreactor system. The needed 10- to 50-fold increase in this yield can be generated by manipulation of the biosynthetic pathway through addition of precursors and elicitors (Shishodia et al. 2008).

Quantification of guggulsterone

Identification of high guggulsterone yielding ecotypes of guggul can pave the way for multiplication and mass propagation of its germplasm through tissue culture and vegetative propagation methods, as well as, selection of suitable geographical locality for its large-scale cultivation or towards regeneration and afforestation of the degraded habitats in nature. This estimation can be done by any chromatographic or spectrophotometric method like Thin layer chromatography (TLC), High performance thin layer chromatography (HPTLC), High performance liquid

Table 5 Application of various biochemical techniques in guggulsterone analysis

Material	Tools	Achievements	References
Commiphora mukul Engl.			
Standard guggulsterone mixture (containing E- & Z-isomers)	HPTLC	Examined the effect of impurities and various degradation conditions (photodegradation, wet heat degradation, acidic and alkaline hydrolytic conditions, and oxidative stress) on the stability of guggulsterone. Provided a powerful tool for elucidating the stability of commercially available herbal drugs and products under the ICH guidelines	Agrawal et al. (2004a)
Guggul capsules and extracts	HPTLC	Estimated the amount of Z-guggulsterone in herbal extract (75 %) and pharmaceutical dosage (65 %)	Agrawal et al. (2004b)
Ethyl acetate extract of commercial products	Reverse phase column HPLC	Identified a suitable analysis method for assaying the active ingredients *E*- and *Z*- guggulsterones, present in commercial guggul resin based products and tablets. Found that the amount of both the guggulsterones is quite lower than the claimed quantity	Mesorb et al. (1998)
Extract of drug-free rat serum	HPLC	Quantified two isomeric forms of guggulsterone (E and Z), simultaneously by HPLC (a C_{18} column using methanol and acetonitrile), after administration of a single dose (50 mg/kg)	Verma et al. (1998)
Commiphora wightii (Arn.) Bhandari			
Ethyl acetate extract of dried stem bark	HPLC	Revealed that there is a great variation in the amount of Z- Guggulsterone (ranges from 0.01 to 0.24 %) among different accessions collected from Gujarat belt. Also found a great variability in Z-guggulsterone in males, females and the hermaphrodites	Gajbhiye et al. (2011)
Plants	HPLC	Evaluated the effect of seasonal and geographical variations on guggulsterone content in germplasm (56 plants) collected from 11 different geographical locations of Rajasthan	Soni et al. (2009)
In vitro and in vivo tissues	Reverse phase column HPLC	Fruit wall and stem (in vivo) contain more guggulsterone (2–4 fold) as compared to zygotic embryos (in vitro). Non-embryonic callus cultures have lowest amount of guggulsterone	Dass and Ramawat (2009c)

chromatography (HPLC) or UV spectrophotometry (Table 5). In case of *C. wightii* very few methods are available for quantification of guggulsterone in pharmaceutical dosages and in biological fluids. Along with this study, it is essential to assess the impurities and stability of drugs in commercialized Ayurvedic extracts and capsules. International Conference on Harmonization (ICH) issued the parent drug stability test guidelines (Q1A). According to these guidelines, analytical assay for testing stability of samples should be completely valid and stability indicating (ICH 1993).The purpose of this study is the assessment of impurities and stability in dosage forms in relation to pure drug. Agrawal et al. (2004a) quantified the standard drug using HPTLC, which separate the drug from its degredative products according to their peaks (different R_F values) on chromatogram. Different peaks were found in various forms of induced stress like in acid (5 N HCl, 80 °C), base (5 N NaOH, 80 °C), oxidative stress (50.0 %), dry heat, wet heat and in photochemical degradation according to their R_F values. These peaks were compared with standard peaks of E- and Z-guggulsterone. Analysis showed that the degraded products had lower R_F values compared to the standard, indicating that degraded products are more polar than the standard analytical compound. During assessment of impurities, two peaks at R_F values of 0.54, 0.70, were found that didn't match with those of marketed capsules, and were considered as impurities which appeared due to mishandling during processing and storage. In another study, Agrawal et al. (2004b) determined the concentration of E- (*Rf* 0.38) and Z- (*Rf* 0.46) guggulsterone in pharmaceutical dosage forms.

The content of guggulsterone is also affected by seasonal (collection time of guggul) and geographical (different rainfall and temperature) variations. For evaluation of these variations, Soni et al. (2009) collected the germplasm (56 plants) from 11 different geographical locations of Rajasthan, India. After analysis of samples by HPLC (for each month from April to March) surprising results were obtained in the amount of guggulsterone with respect to collection timing and locations. Guggulsterone content was significantly higher during summer (May–July, highest in May) and gradually decreased in the rainy seasons (Aug–Oct) and was lowest in winter (Nov–March) because of dormant stage of plants, indicating that May–July months are ideal for collection of guggulsterone. With reference to the geographical locations northern, western and central part of Rajasthan showed maximum amount (1.87–2.76 %, highest in rocky tracts of Mangliawas, National Herbal Farm, Ajmer) whereas southern part of the state produced lower amount (0.61–1.19 %, lowest in Pratapgarh) of guggulsterone. Among all the samples, Z isomeric form was dominating over E-guggulsterone. They also depicted a strong relationship between average rainfall of the area and guggulsterone content. Highest (>1.87 %) and lowest (<1.63 %) guggulsterone was estimated in areas having rainfall between 15 and 55 cm and more than 60 cm respectively.

By using reverse phase column HPLC in cell and callus cultures of *C. wightii* high guggulsterone content can be harvested (Dass and Ramawat 2009c). Analysis showed that fruit wall and stem (in vivo) of *C. wightii* contain more guggulsterone (2–4 fold) as compared to zygotic embryos (in vitro), whereas non-embryonic callus cultures have lowest amount of guggulsterone. Therefore, embryogenic callus is the best alternate for guggulsterone production in vitro (Kumar et al. 2006). Quantification of E- and Z-guggulsterone (in blood plasma) by HPLC was reported by Singh et al. (1995). Verma et al. (1998) quantified two isomeric forms of guggulsterone (E and Z), simultaneously by HPLC (a C_{18} column using methanol and acetonitrile), in rat serum after administration of a single dose (50 mg/kg). It was of interest to note that the chromatograms of serum samples from dosed rat showed an additional major peak at a retention time of 5.50 min. This peak was identified on the basis of its UV spectrum and retention time (E-isomer). Repeated analysis of analytical standards and spiked serum samples during the assay validation showed that the Z to E conversion does not take place in the spiked serum samples and hence the formation of the E-isomer can be attributed solely due to the in vivo process.

HPLC method for quantification of E and Z stereoisomers with highest recovery (99.5 %) and precision was validated by Mesorb et al. (1998) in oleo-gum resin exudates of *C. mukul* along with its products (capsules). This process was validated for quantification of guggulsterone E and Z (15–85 and 25–130 µg, respectively) on reversed phase C_{18} column on acetonitrile–water gradient. After quantification of six commercial products, it was found that all the products contained significantly less amount of this herbal compound than their claimed amount. The amount of Z-guggulsterone estimated in herbal extract and in pharmaceutical dosage were 75 % (w/w) and 65 % (w/w) respectively while in standard guggulsterone mixture it was identified as only 45 % (w/w).

Genetic diversity assessment

Genetic variation is important because it provides the "raw material" for natural selection. For conservation of this endangered species it is necessary to understand the nature of population structure (Haque et al. 2009a). Characterization of germplasm is essential for identifying individual genotypes as well as to access the extent of variability existing among the accessions. These are considered as valuable tools for plant breeding programs as well as for evolutionary and conservation studies. Therefore, it is necessary to detect and document the amount of variation existing within and between the populations. The comprehensive information obtained from such an exercise would help breeders, geneticists and conservationists to effectively utilize the valuable genetic resources (Kalia et al. 2011). The genetic diversity studies will aid the breeder to assess the variability of the traits of importance and to choose the parents for hybridization program. A broad genetic base can ensure long-term survival of a species because negative traits such as inherited diseases can become widespread within a population when that population is left to reproduce only within its own members. Some of the important investigations related to *Commiphora* germplasm characterization using various markers are summarized in Table 6.

Table 6 Characterization and diversity assessment of *Commiphora* germplasm

Technique	Achievements	References
Morphological markers		
Resin texture, leaf, pseudo-aril, putamen	Described a new species *Commiphora buruxa* Swanepoel, closely related to *C. cervifolia* J.J.A.van der Walt, from Gariep Centre of Endemism	Swanepoel (2011)
Fruits, seeds, phytosociology	Investigated various morphological parameters of guggul plants growing under various natural habitats of western India	Lal and Kasera (2010)
Biochemical markers		
Chemical markers and metabolites	Checked the authenticity of a commercial guggul sample and found *Mangifera indica* gum, as an adulterant	Ahmed et al. (2011)
Essential oil profiles	Reported that the four *Commiphora* species viz : *C. ornifolia*, *C. parvifolia*, *C. planifrons* and *C. socotrana* endemic to the Soqotra Archipelago of Yemen can be distinguished from myrrh trees of other places based on essential oil profiles	Mothana et al. (2009)
Essential oil profiles	Differentiation can be done between *C. erythraea* and *C. kataf*, the two often confused species	Marcotullio et al. (2009)
Enzyme systems	Analyzed crude enzyme extracts of 22 accessions of *C. wightii* to determine allele frequency, Polymorphic loci and genetic distance	Kalpesh and Mohan (2008)
Molecular markers		
RAPD, ISSR	Used 60 RAPD primers and 27 ISSR primers to study genetic relationships among 24 genotypes of *C. wightii*, collected from different parts of Rajasthan	Samantaray et al. (2011)
RAPD	Analyzed the diversity among 7 populations of *C. wightii* collected from different parts of Rajasthan (India)	Haque et al. (2009b)
RAPD	Tried to identify sex-specific molecular markers in dioeceious and hermaphrodite plants of *C. wightii*	Samantaray et al. (2009)
RAPD	Evaluated genetic relationship among different aceessions collected from Rajasthan and Gujarat states of India using a set of 40 primers	Suthar et al. (2008)
ETS, ITS sequences	Conducted molecular phylogenetic analysis of *Commiphora*, predominantly growing in tropical Africa	Weeks and Simpson (2007)
ITS sequences	Concluded that *C. wightii* plants have evolved under reproductive isolation, probably due to population fragmentation and habitat loss	Haque et al. (2009a)

Morphological markers

Traditionally, diversity within and between populations is determined by assessing differences in morphology. However, morphological determinations need to be taken by an expert in the species as they are subject to changes due to environmental factors and may vary at different developmental stages. Moreover, the morphological characters are limited in number (Kalia et al. 2011). *Commiphora buruxa* Swanepoel, described as a new species, is known only from the Gariep Centre of Endemism, southwestern Namibia (Swanepoel 2011). It appears to be most closely related to *C. cervifolia* J.J.A. van der Walt. Diagnostic morphological characters of *C. buruxa* include a viscous, cream-coloured exudate, variably simple to trifoliolate leaves, and a putamen covered by a small copular pseudo-aril. The species is known from less than 20 plants. Lal and Kasera (2010) investigated various morphological parameters of fruits and seeds and phytosociology (vegetative dynamics) of guggul plants growing under various natural habitats of western India. The percentage of black and white seeds was 38 and 62 respectively under field collections. Black seeds showed 60 % viability and 40 % germination under nursery conditions.

Biochemical markers

Morphologically, it is not easy to distinguish guggul oleo-gum resin from gum resins of other plants. Reliability of the commercially available guggul is critical keeping in view its wide use in pharmaceuticals and food products. Adulteration can lead to dangerous consequences. To overcome the limitations of morphological traits other markers have been developed at both biochemical and molecular level.

To check the authenticity of a commercial guggul sample, chemical markers and 17 metabolites were identified, including three new 20(S),21-epoxy-3-oxocholest-4-ene (1), 8 β-hydroxy-3,20-dioxopregn-4,6-diene (2), and 5-(13′ Z-nonadecenyl) resorcinol (17) from the ethyl acetate soluble part by Ahmed et al. (2011). During this study, four compounds were identified as constituents of *Mangifera indica* L. gum, as an adulterant in the commercial guggul sample. This discovery highlighted the common malpractices in the trade of medicinal raw material in the developing world. The structures of the compounds were deduced by the spectroscopic technique and chemical methods, as well as by comparison with the reported data. The structure of 20(S), 21-epoxy-3-oxocholest-4-ene (1) was deduced by single-crystal X-ray diffraction.

The essential oil of *C. erythraea* Engl. is rich in sesquiterpenes, particularly furanosesquiterpenes (50.3 %). GC–MS analysis of the oil permitted differentiation between *C. erythraea* Engl. and *C. kataf* Engl., two often confused species (Marcotullio et al. 2009). Four species of the genus *Commiphora* viz : *C. ornifolia* (Balf. f.) J. B. Gillett, *C. parvifolia* Engl., *C. planifrons* Engl. and *C. socotrana* Engl. are endemic to the Soqotra Archipelago of Yemen, known as the biodiversity "jewel" of the Arabian Sea. These species are most important for the Soqotri folk and are used to make naqf (from bark and under bark) to cure wounds and ulcers in both humans and livestock**.** Mothana et al. (2009) analyzed the essential oil of these species obtained by hydrodistillation. They reported that the chemical profiles of *Commiphora* oils as examined by GC and GC–MS could be used as a chemo-taxonomical marker to distinguish between the myrrh-trees on Soqotra Island and other places.

Kalpesh and Mohan (2008) analyzed crude enzyme extracts of 22 accessions of *C. wightii* using nine enzyme systems to determine the allele frequency, Polymorphic loci (P), Allele number (A), Shannon's information index (I), Heterozygosity and homozygosity, genetic distance and identity. These enzyme systems provided a total of 13 loci and 31 alleles for 22 accessions of *C. wightii*. Average observed (Ho) and expected heterozygosity (He) showed moderate levels of genetic variation among different accessions. The genetic identity (I) was observed highest between accessions IC-370 and IC-378 and lowest between accessions IC-389 and IC-377. Dendrogram based on UPGMA clearly depicted the spectra of genetic diversity among various accessions and formed into two major groups and divided into six clusters. Nei's genetic distance was found to be highest between the accessions IC-389 and IC-375 and minimum between the accessions IC-383 and IC-380. Based on the results of this study, they suggested that the accessions IC-370 and IC-373 should be conserved and maintained in the field gene bank for future use.

Molecular markers

During the past decades, classical methods to evaluate genetic variation have been complemented by molecular techniques. These are useful for characterizing the genetic diversity among different cultivars or species, for identifying genes of commercial interest and improvement through genetic transformation technology (Kalia et al. 2011). Since variability is a prerequisite for selection programme, it is necessary to detect and document the amount of variation existing within and between the populations.

A lot of work has been done on pharmaceutical applications of guggulsterone, a bioactive compound of *C. wightii,* but genetic diversity is less characterized (Table 6). Genetic characterization is helpful in understanding the various intrinsic factors which are responsible for production of secondary metabolites, thus study can helpful in commercialization of any economically and pharmaceutically important species. Molecular markers have been widely used for the genetic characterization of germplasm of several cultivated species. They have been utilized for a variety of purposes including construction of linkage maps, examining genetic relationships between individuals, and identification of crop cultivars. Very limited molecular studies have been done on characterization of guggul at national and international level.

Haque et al. (2009b) used 100 RAPD primers to find the genetic variation among 7 populations of *C. wightii*. Of the 100 random primers screened, 44 primers yielded 220 loci. Statistical analysis indicated low genetic diversity ($H_{pop} = 0.0958$; $I = 0.1498$; mean polymorphic loci $= 14.28$ %), and high genetic differentiation among the populations ($G_{ST} = 0.3990$; AMOVA Φ_{ST} of 0.3390; Bayesian $\theta^{(II)} = 0.3002$). The low genetic diversity was attributed to geographic isolation and restricted gene flow ($N_m = 0.7533$) between the fragmented populations. Molecular phylogenetic analysis of *Commiphora*, predominantly growing in tropical Africa provided an insight into the evolution and historical biogeography of this genus which was described as an "impossible" genus (Weeks and Simpson 2007). The crown group radiation of *Commiphora* and its sister genus Vietnamese *Bursera tonkinensis* Guillaumin was found to correspond with the onset of the Miocene.

Samantaray et al. (2011) used 60 RAPD primers and 27 ISSR primers to study genetic relationships among 24 genotypes of *C. wightii*, collected from different parts of Rajasthan. Cluster analysis, using UPGMA, SHAN clustering (NTSYS-pc) grouped the genotypes into three main clusters which were further subdivided into six sub-clusters in RAPD analysis while ISSR showed four main clusters and nine sub-clusters. The genetic similarity among the genotypes varied from 0.43 to 0.97 when pooled RAPD and ISSR data were used. Hence the study showed existence of wide genetic variability within the species which is otherwise reported as apomictic species and opened the scope of selection of better genotypes in relation to chemical data.

Based on sequences of ITS region of nuclear ribosomal DNA of 32 individuals, Haque et al. (2009a) concluded that *C. wightii* plants have evolved under reproductive isolation, probably due to population fragmentation and habitat loss. The reproductive isolation, coupled with the reported apomictic behaviour may have a detrimental effect on genetic variation. Correlation between genetic and geographic distance was statistically significant and it appeared that present population is a remnant of a previously large population that was present in the area. Since population differentiation is high, the population continuum has been disrupted, possibly due to overexploitation, unsustainable utilization and other anthropogenic activities.

RAPD primers were also used to identify sex-specific molecular markers in dioeceious and hermaphrodite plants of *C. wightii*. Sixty RAPD primers were screened out of which only three primers were found to be associated with sex expression. A $\sim$1,280-bp fragment from the primer OPN06 was found to be present in all the female individuals. Another primer OPN 16 produced a unique $\sim$400-bp amplification product in only hermaphrodite individuals. The third marker, OPA20 amplified a $\sim$1,140-bp fragment from female and hermaphrodite DNAs, but failed to do so from the DNA of male plants (Samantaray et al. 2009).

However, no markers have been identified till date which are linked to guggulsterone content or sex in *Commiphora*, both of these characters are very important from conservation point of view as the species is valued for its oleo-gum resin and is also dimorphic.

Molecular basis of guggulsterone action

The constituents of oleo-gum resin guggul, primarily guggulsterone E and Z, have been reported to have an array of pharmacological applications including lipid and cholesterol lowering properties. Recently, reports regarding its role as an anti-cancerous agent have also been published. However, the physiological and molecular basis of such pharmacological effects needs to be pin pointed. Reports have shown that guggulsterone modulate many targets including growth factors, growth factor receptors, transcription factors, cytokines, enzymes and genes regulating apoptosis. Shishodia et al. (2008) have reviewed the molecular targets of guggulsterone in detail.

Guggulsterone has been shown to induce apoptosis and suppress proliferation, invasion, angiogenesis and metastasis of tumor cells. Various mechanisms have been suggested to explain the anti-carcinogenic effects of guggulsterone, including inhibition of reactive oxygen intermediate (ROI), suppression of inflammation and inhibition of nuclear receptors, transcription factors, inflammatory cytokines, antiapoptotic proteins, cell survival pathways, cyclo-oxygenase (COX-2), matrix metalloproteinase 9 (MMP-9), inducible nitric oxide synthase (iNOS) and cell cycle related proteins. Both E and Z forms of guggulsterone display a high affinity for steroid receptors, including androgen, glucocorticoid, and progesterone receptors. Guggulsterone mediates gene expression through regulation of various transcription factors and steroid receptors. Evidence has been presented to suggest that guggulsterone can suppress tumor initiation, promotion and metastasis (Shishodia et al. 2008).

Guggulsterone has been described as a farnesoid X receptor (FXR) antagonist and it has been suggested that inhibition of FXR activation may be the basis for the cholesterol-lowering activity of guggulsterones (Urizar et al. 2002). Deng (2007) reported that enhanced bile salt export pump (BSEP) expression by guggulsterone represents a possible mechanism by which guggulsterone exerts its anticancer and hypolipidemic effects. Guggulsterone may operate through suppression of a nuclear transcription factor kappaB (NF-κB) which is required for expression of genes involved in cell proliferation, cell invasion, metastasis, angiogenesis and resistance to chemotherapy (Shishodia and Aggarwal 2004). Guggulsterone suppressed cytokine-induced expression of inducible nitric oxide synthase (iNOS*)* mRNA and protein. These observations suggest that guggulsterone may have anti-metastatic activity through suppression of iNOS (Lv et al. 2008). Most inflammatory stimuli are known to activate three independent mitogen activated protein kinase (MAPK) pathways (Shishodia et al. 2007). Singh et al. (1994) showed that guggulsterone-induced cell death in human prostate cancer cells is caused by reactive oxygen intermediate (ROI)-dependent activation of one of the MAPK pathways. Guggulsterone has also been shown to modulate several proteins involved in regulation of the cell cycle. Cyclin D1 has been shown to be overexpressed in many types of cancer, including those of the breast, esophagus, head and neck, and prostate (Caputi et al. 1999). It is possible that the anti-proliferative effects of guggulsterone are due to inhibition of cyclin D1 expression (Shishodia et al. 2007).

Guggulsterones E and Z are responsible for the lipid lowering properties of guggul in human blood and at least four mechanisms have been proposed to explain their activity (Ramawat et al. 2008; Shishodia et al. 2008). First, guggulsterones might interfere with formation of lipoproteins by inhibiting biosynthesis of cholesterol in the liver (Gupta et al. 1982). Second, guggulsterones have been shown to enhance the uptake of LDLs by the liver through stimulation of LDL receptor–binding activity in the membranes of hepatic cells (Singh et al. 1990). Third, guggulsterones increase fecal excretion of bile acids and cholesterol, substantially decreasing the rate of absorption of fat and cholesterol in the intestine (Gupta et al. 1982). Finally, guggulsterones directly stimulate the thyroid gland (Tripathi et al. 1984). Several animal studies and clinical trials have been performed to evaluate the hypolipidemic effects of guggul.

Although guggulsterone was shown to inhibit proliferation and induce apoptosis in cancer cells, further preclinical and clinical studies are required to establish the anti-cancer potential of guggulsterone. The mechanism of action of guggulsterone has begun to reveal new uses of this drug. The bioavailability, pharmacokinetics, pharmacodynamics and metabolism of guggulsterone require further examination (Shishodia et al. 2008).

Conclusions and future prospects

Commiphora wightii is pharmacologically, economically and ecologically important endangered plant species, occurring in India, Bangladesh and Pakistan. Increasing demand coupled with its unscrupulous exploitation has created a serious problem as demand is out pacing the supply which has endangered the survival of this species in the wild. Although this plant has many excellent traits it is still in an early phase of domestication. No efforts have been made to improve the taxon through selection and conventional breeding programs. Some issues which need immediate attention include multiplication and conservation, development of plantations, improved scientific tapping processes, disease and pest resistance, production of guggulsterone in vitro, development of guggulsterone and sex linked markers, and genetic diversity analysis among others.

Being an endangered species there is a need to address the issues of multiplication and conservation more rigorously. This will require the application of micropropagation techniques for production of planting material in bulk quantities. The available protocols have demonstrated the potential of micropropagation for multiplication, however, it has many limitations including slow growth, heavy contamination of field derived explants, low rooting rates as well as field establishment rates. Similarly, somatic embryogenesis has limitations like asynchronous development and low conversion rates. Difficulty in dissecting the hard fruits to obtain ovules, low frequency of embryogenic responses of zygotic embryos, explant yellowing and death during culturing and genotypic differences among cultivars during culture establishment in vitro are the additional constraints.

Commiphora wightii is vulnerable to various types of pests, pathogens and climatic constraints during cultivation like other woody perennials. Therefore, development of genetically engineered plants capable to counter such biotic and abiotic stresses is imperative. However, an efficient regeneration protocol must be in place before genetic transformation studies can be initiated. In addition, there is a need to understand the guggulsterone biosynthetic pathway in detail so that transgenics with better guggulsterone levels can be created. Use of different precursors and elicitors to obtain high gugulsterone content in cell and callus cultures will also help in alleviating the pressure on natural guggul population. Therefore in vitro regeneration and secondary metabolite production techniques need to be improved further.

Detailed information must be in hand regarding the genetic makeup and mode of propagation of *Commiphora* under natural conditions before any selection and breeding programs can be initiated. Apomictic behaviour along with reproductive isolation has a negative effect on genetic variation (Haque et al. 2009a). Availability of a broad genetic base is must for initiating breeding programmes in any given crop. There is a need to characterize the germplasm to identify superior germplasm with respect to guggulosterone content so that efforts can be directed towards its conservation and multiplication. Diversity studies have been initiated in localized pockets in *Commiphora* growing regions of the world using RAPD and ISSR markers which need to be strengthened further so that a genetic base for breeding and transgenic applications may be prepared. The available genetic base can be broadened using modern tools of biotechnology including in vitro selection of somaclonal variations, mutagenesis and transgenics. Knowledge of genetic relationships in parental varieties could improve the effectiveness of breeding programmes. Robust markers need to be in place for determining the sex of the seedlings/plants. There is ample scope for research on the identification of sex and development of guggulsterone linked markers in *C. wightii*.

Commiphora grows under very harsh agro-climatic conditions in the arid regions therefore, must be harbouring many important genes conferring abiotic stress tolerance to the species. Therefore, the germplasm must be conserved and analyzed properly so that these abiotic stress tolerance genes may be identified and harvested for use in other economically important crops and trees in the future.

A high medicinal and therapeutic value of the oleo-gum resin of the plant has led to uncontrolled ruthless exploitation of *C. wightii* resources in areas of its natural distribution. Thus, protection and conservation of the valuable germplasm have become urgent. Mangliawas, Ajmer is the only National Herbal Farm where thousands of cutting raised female *C. wightii* plants has been established. Conservation efforts will require the involvement of local people and villagers. Additional efforts are also required to elucidate scientific tapping practices so that the plant is not killed after tapping. Recently, 'Save guggul project' has been started in Rajasthan with the involvement of local tribal populations. A series of awareness programs are being carried out, under this project, in various parts of Rajasthan to make people aware about the importance and conservation of guggul plants. In such conditions, conservation of agronomically important cultivars through in vitro methods and cryopreservation must be done. It will be the most promising answer to biological and climatic hazards which may threat the germplasm maintained *in situ* in field genebanks and germplasm gardens. Cryopreserved material (stored as seeds, ovules, embryos, callus, etc.) can be used successfully for breeding in the future (Kalia et al. 2011).

Commiphora wightii has become a vital subject of research due to the presence of numerous pharmaceutically valuable bioactive agents. Although no other natural product is as effective as guggulsterone against treatment of multiple ailments including lipid- and cholesterol lowering activity but there is a need to evaluate the complexity of different bioactive components present in the oleo-gum resin by rigorous assessment of the mechanism of drug action, pharmacokinetics, metabolism, effective concentration, ratio of impurities, bioavailability and bibliographic data which will reveal the safety and efficacy of the drug after a long-term experience. Therefore, cost effective and easily available drug delivery systems like guggulsome having no toxicity can be established on a large scale. Molecular mechanisms underlying guggulsterone action must be elucidated precisely so that various pharmaceutical applications of guggulsterone including anti-cancerous and cardio-protective properties can be used for saving thousands of lives annually.

There is no doubt that the future holds great promise for this endangered genus, *Commiphora wightii*. This ancient plant with its powerful and healing synergies has much to contribute to this planet and its inhabitants. The environmental problems being faced in global arid and semi-arid regions are very serious and need to be addressed urgently. *C. wightii* has the capacity to improve the environment and economic development of the regional population if systematic reforestation programs are undertaken in the region. We can look forward to a continued revelation of *C. wightii*'s many gifts through the increasing interest and research into its abundant and valuable properties. Judicious exploitation and utilization of *Commiphora wightii* resources can bring more benefits to mankind throughout the world.

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